DATA 6550: Visualization Ethics and Communication

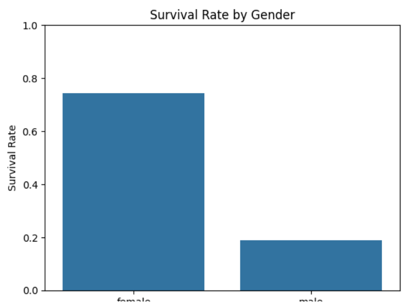
Titanic Dataset Analysis: (The good and Bad Visualization)

1. Introduction : Exploring moral and immoral visualization ways utilizing the Titanic dataset was the purpose of this research. Within the Titanic dataset you will find the demographics and survival statistics of the 891 people who were passengers aboard the RMS Titanic. Several parameters were explored, including gender, age, fare, embarkation port, and passenger class, in order to identify surprising survival trends. After gathering critical insights, we developed both honest and purposefully deceptive visuals to show how design choices effect interpretation.

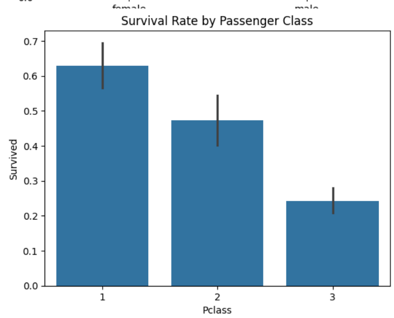
2. Dataset Overview : Eight hundred ninety-one persons and twelve variables encompassing passenger class, gender, age, fare, embarkation port, and survival status constitute the Titanic training dataset. We checked over the dataset and found any missing values before producing visualizations. The Age field had 177 missing data, the Cabin column had 687, and the Embarked column had only 2 items. Most passengers were in third class, and just over 38 percent made it to their destination alive. Several significant trends were identified using the exploratory analysis:

- Compared to male travelers, female passengers were more likely to survive.
- Compared to third-class guests, first-class travelers had considerably superior survival rates.
- Each embarkation port has a varied survival rate.
- Passengers' economic inequalities were clear in the very uneven fare system.
- We were able to pick the variables for our visualizations owing to these findings.

3. Accurate graphics: Our initial aim was to produce pictures that accurately and clearly represent information.



3.1 Gender-Based Survival : Male and female passengers' survival rates were compared in the first image. The survival percentage for female passengers was 74.2% (233 out of 314), but the survival rate for male passengers was merely 18.9% (109 out of 577). The comparison was easy to read thanks to the chart's continuous scale, clear labeling, and precise colors.



3.2 Passenger Class Survival : The survival statistics for each passenger class were shown in the second image. Passengers in first class had the best survival rate (63.0%), followed by those in second class (47.3%) and third class (24.2%). Without overstating the disparities, the image strongly underscored the relationship between survival and socioeconomic position. Because they transmit comprehensive information, use consistent scales, and allow viewers to correctly interpret the results, these visualizations comply to ethical visualization principles.

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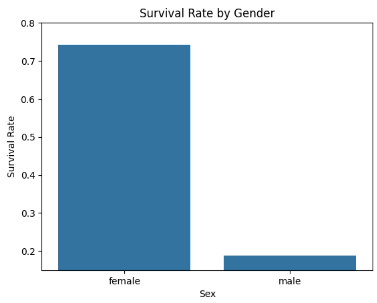
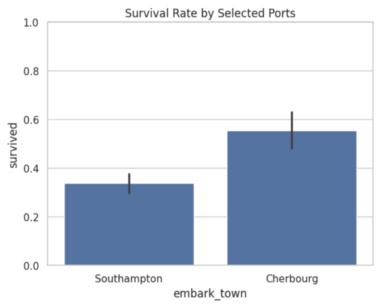
4. Deceptive Visuals : We purposely made fraudulent charts with the same information in order to acquire visualization ethics.



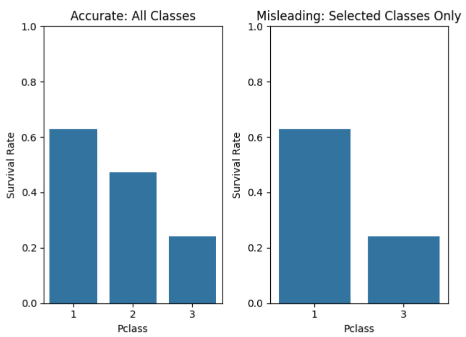
4.1 Y-axis truncation : Survival rates by embarkation port were shown in the first misleading presentation, except the y-axis started at 0.30 rather than 0. The shorter axis made the disparities appear much wider than they actually were, even though the actual survival rates were about 33.7% (Southampton), 38.9% (Queenstown), and 55.3% (Cherbourg). This demonstrates how perception may be affected by only adjusting the axis without affecting the underlying facts.

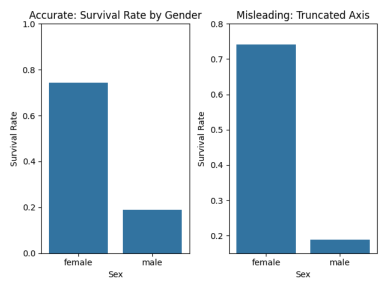


4.2 Selecting Particular Data : Queenstown was purposely omitted out of the second false photo, which only compared Southampton with Cherbourg. The graphic increased the apparent gap between the remaining ports and presented an incomplete comparison by removing one category. This example explains how selective data selection may mislead viewers and result in inaccurate conclusions. These pictures explain how even when the data itself doesn't change, representations may become deceptive.

5. Moral Aspects : The concept that visualization design entails ethical responsibility is among the project's most fundamental lessons. People's viewpoints of information may be greatly changed by activities like altering axis scales, omitting out categories, or accentuating specific facts. Accurate data display, proper labeling, proportional scale, and the avoidance of hiding vital context are all components of ethical visualizations. In domains including healthcare, business, economics, and public policy, where inaccurate interpretations could have devastating effects, misleading representations may have an influence on decisions. This analysis indicated that when conveying data-driven notions, honesty and transparency are vital.

6. Conclusion : Because survival was influenced by a multiplicity of demographic and socioeconomic characteristics, the Titanic dataset gives an excellent case study for analyzing visualization ethics. We proved via exploratory inquiry that the largest predictors of survival were passenger class and gender. We emphasized how small design changes may drastically influence a viewer's perception without altering the fundamental information by compared ethical versus misleading infographics. The utility of developing images that correctly, equitably, and ethically portray information was proven by this attempt.





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